

Robust and Efficient Hybrid Functional Maps for Non-Rigid Shape Matching

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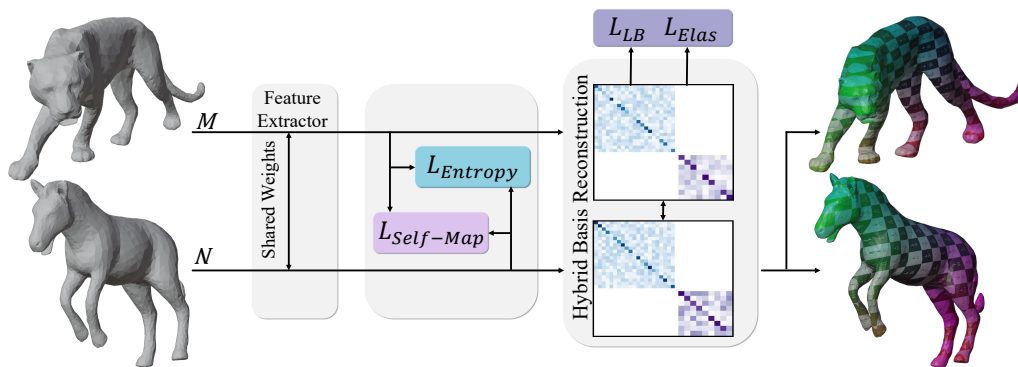


Figure 1: Overview of the pipeline of our unsupervised method. First, features are extracted using a shared-weight feature extractor and regularized with $L_{\text{self-map}}$ and L_{entropy} . These features are then projected onto basis sets from different operators to reconstruct the functional map in the hybrid function space. Finally, L_{LB} , L_{Elas} , and additional regularization terms impose constraints on the hybrid functional map.

Abstract

In the deep functional map framework for non-rigid shape matching, combining the intrinsic Laplace-Beltrami operator (LBO) with the elastic thin-shell Hessian captures global deformations and high-frequency variations like bending and creases. However, the elastic basis requires solving a $k^2 \times k^2$ system, leading to high computational cost and potential instability. We propose a novel unsupervised method that reconstructs functional maps in a hybrid spectral space, improving training stability and speed. By analyzing the point-wise map relationships and introducing two new loss functions, we alleviate the ambiguity in the mapping. Extensive experiments demonstrate that our method outperforms previous approaches as well as some of the latest state-of-the-art methods.

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CCS Concepts

• Computing methodologies → Shape analysis; Computer vision.

Keywords

shape matching, functional maps, geometry and modeling

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1 Introduction

The establishment of correspondences between 3D shapes is a crucial problem in shape analysis, computer vision, and computer graphics. Despite decades of research [Sahillioğlu 2020], accurately determining point-wise correspondences remains challenging, especially with large-scale non-isometric deformations or topological noise.

With deep learning, data-driven approaches, particularly the functional map framework by Ovsjanikov et al. [Ovsjanikov et al.

2012], have become dominant. The eigenfunctions of the LBO are invariant under isometries, capturing low-frequency shape information, making them efficient for optimization and robust to noise. However, low-frequency spectral mapping often results in local detail inconsistencies.

Hartwig et al. [Hartwig et al. 2023] applied functional maps using a basis from elastic thin-shell energy spectral decomposition, with modes sensitive to extrinsic surface curvature. Bastian et al. [Bastian et al. 2024] further developed a hybrid basis combining LBO and thin-shell Hessian energy, addressing earlier limitations. However, computing functional maps in this hybrid basis requires solving a $k^2 \times k^2$ system, imposing significant computational costs and potential instability, which leads to gradient explosion and training failure.

We propose a novel unsupervised shape matching method to address these shortcomings. Our approach avoids directly computing functional maps on a hybrid basis by using learned descriptors, eliminating the need for the complex $k^2 \times k^2$ system. Additionally, we analyze the underlying conditions of point-wise maps and introduce an entropy loss and a self-map loss to enable more accurate correspondences. We summarize our primary contributions as follows: (1) We propose a novel unsupervised hybrid method that significantly avoids the instability observed in existing methods during training. Our approach becomes more efficient and simpler by leveraging the advantages of basis functions from different operators. (2) We provide a theoretical analysis of the point-wise maps based on feature similarity and introduce self-map loss and entropy loss to alleviate the ambiguity in the mapping process. (3) Across various challenging benchmarks, including isometric and non-isometric matching scenarios and cases involving topological noise, our method outperforms the latest supervised methods and achieves state-of-the-art performance.

2 Method

Our goal is to estimate robust functional and point-wise maps for shapes. While the hybrid basis improves high-frequency detail alignment, it incurs high computational costs due to the $k^2 \times k^2$ Hilbert-Schmidt(HS) norm expansion, causing gradient explosion from large elastic functional map values. We propose a novel method that constructs functional maps on a hybrid basis, reducing computational complexity. Fig. 1 illustrates our framework. The following sections cover hybrid basis reconstruction (Sec. 2.1), map analysis (Sec. 2.2), and overall losses (Sec. 2.3).

2.1 Hybrid Basis Reconstruction

The hybrid basis approach [Bastian et al. 2024] decomposes the optimization into two separate problems, while the energy term is expanded into a $k^2 \times k^2$ linear system and is solved as:

$$(A^\top A + \lambda \zeta^\top \zeta) \text{vec}(C) - A^\top \text{vec}(B) = 0, \quad (1)$$

where:

$$A = (\sqrt{M_{k,2}} D_{\Psi_1})^\top \otimes I, \quad B = \sqrt{M_{k,2}} D_{\Psi_2},$$

$$\zeta = (\Lambda_1 \sqrt{M_{k,1}^{-1}}) \otimes \sqrt{M_{k,2}} - \sqrt{M_{k,1}^{-1}} \otimes (\sqrt{M_{k,2}} \Lambda_2),$$

and $\otimes$ denotes the Kronecker product.

However, due to the complexity of the $k^2 \times k^2$ expansion, the computational cost becomes prohibitive as the number of elastic basis functions increases. To address this issue, by solving the optimal transport problem, we can construct a point-wise map:

$$\arg \min_P \|F_M - P F_N\|_F^2. \quad (2)$$

The orthogonal projector operator [Bastian et al. 2024] for the hybrid basis is given by:

$$\Theta_i^\dagger := M_{k_{\text{total}},i}^{-1} \Theta_i^\top M_i. \quad (3)$$

The off-diagonal blocks in the reduced mass matrix can be assumed to be zero; a more detailed explanation is provided in HybridFmap [Bastian et al. 2024]. Thus, the orthogonal projector can be written in block form:

$$\Theta_i^\dagger = \begin{bmatrix} \Phi_i^\top M_i \\ M_{k_{\text{Elas},i}}^{-1} \Psi_i^\top M_i \end{bmatrix} = \begin{bmatrix} \Phi_i^\dagger \\ \Psi_i^\dagger \end{bmatrix}. \quad (4)$$

We then directly construct the functional map representation by utilizing the point-wise map, with the following formulation:

$$C = \Theta_i^\dagger P \Theta_j. \quad (5)$$

This approach eliminates the need to solve a complex $k^2 \times k^2$ system. Instead, the functional map can be obtained through simple matrix multiplication on a hybrid basis. Ultimately, the functional map is expressed as:

$$C = \begin{pmatrix} \Phi_i^\top M_i P \Phi_j & \Phi_i^\top M_i P \Psi_j \\ M_{k_{\text{Elas},i}}^{-1} \Psi_i^\top M_i P \Phi_j & M_{k_{\text{Elas},i}}^{-1} \Psi_i^\top M_i P \Psi_j \end{pmatrix}. \quad (6)$$

Moreover, we impose the following constraint on the above equation:

$$\Phi_i^\top M_i P \Psi_j = M_{k_{\text{Elas},i}}^{-1} \Psi_i^\top M_i P \Phi_j = 0. \quad (7)$$

This is because the mapping between the bases is undesirable.

2.2 Map Analysis Based on Feature Similarity

The pairwise point-wise map Π_{NM} is computed by measuring the similarity between features F_N and F_M , and is theoretically a (partial) permutation matrix. The point-wise correspondence, derived from feature similarity, is defined as:

$$\Pi_{NM} = \text{Softmax} \left(\frac{F_N F_M^\top}{\tau} \right). \quad (8)$$

Thus, through Eq. 8, we can also derive Π_{NN} and Π_{MM} . For the permutation matrix Π_{NN} , each row and column should strictly contain exactly one nonzero element. Suppose F_N has duplicate rows, for instance, $F_N[i, :] = F_N[j, :]$. In such cases, we can construct a permutation matrix Π_{NN} that swaps row i with row j while keeping all other rows unchanged. Consequently, there exists a non-identity permutation matrix $\Pi_{NN} \neq I$ such that:

$$\Pi_{NN} F_N = F_N. \quad (9)$$

Similarly, for Π_{MN} , we can construct an equivalent solution as:

$$\Pi'_{MN} = \Pi_{MN} \Pi_{NN}. \quad (10)$$

At this point, different permutation matrices Π_{NN} satisfy the equation in Eq. 9. Each Π_{NN} generates an equivalent solution $\Pi'_{MN} = \Pi_{MN} \Pi_{NN}$. In other words, if Π_{MN} is a solution, multiplying it by

Table 1: The generalization results were on FAUST and SCAPE training and testing and on three data sets, respectively. In most cases, our approach surpasses earlier axiomatic and supervised methods and the most recent unsupervised techniques while showing exceptional generalization performance across different datasets. The best results are marked in bold, while the second-best results are underlined.

Method	FAUST			SCAPE			FAUST+SCAPE			SMAL	DT4D-H		TOPKIDS
	FAU	SCA	SHR	FAU	SCA	SHR	FAU	SCA	SHR		intra-class	inter-class	
Axiomatic													
ZoomOut	6.1	–	–	–	7.5	–	–	–	–	38.4	4.0	29.0	33.7
Smooth Shells	2.5	–	–	–	4.7	–	–	–	–	36.1	1.1	6.3	11.8
Supervised													
FMNet	11.0	30.0	–	33.0	17.0	–	–	–	–	42.0	9.6	38.0	–
GeomFMaps	2.6	3.3	9.9	3.0	3.0	12.2	2.6	3.0	7.9	8.4	2.1	4.1	–
Unsupervised													
AFMap	1.9	2.6	6.4	2.2	2.2	9.9	1.9	2.3	5.8	5.4	1.7	11.6	23.4
DUO-FMNet	2.5	4.2	6.4	2.7	2.6	8.4	2.5	4.3	6.4	6.7	2.6	15.8	–
ULRSSM	1.6	3.6	7.2	1.9	1.9	7.6	1.7	3.2	4.6	4.2	0.9	4.5	9.2
EOT-FMap	<u>1.5</u>	3.4	<u>5.5</u>	<u>1.6</u>	1.8	7.0	1.6	2.2	4.7	4.0	–	4.2	–
HybridFmap	1.4	3.0	<u>6.3</u>	1.5	1.9	5.8	1.5	2.0	4.3	3.7	1.0	3.5	6.6
Our Method	1.4	2.0	4.4	1.7	1.8	3.7	1.4	1.9	3.1	2.7	0.9	3.4	4.9

a non-identity permutation matrix Π_{NN} results in another valid solution Π'_{MN} . This indicates that duplicate rows in F_N imply that certain points in the feature space are indistinguishable, leading to ambiguity in the mapping.

To address this issue, we propose a self-map loss that regularizes the point-wise self-mapping to approximate the identity matrix:

$$L_{\text{self-map}} = \|\Pi_{ii} - I\|_F^2. \quad (11)$$

Meanwhile, we note that for differentiable computation, Eq. 8 utilizes a softmax function to generate a probabilistic correspondence matrix. However, strictly speaking, the correspondence matrix should be a hard matrix, defined as:

$$\Pi \in \{0, 1\}^{n_M \times n_N} : \Pi 1_{n_N} = 1_{n_M}, 1_{n_M}^\top \Pi \leq 1_{n_N}^\top. \quad (12)$$

Ideally, the point-wise map matrix based on feature similarity should closely approximate a hard matrix. To achieve this, we minimize entropy to encourage the mapping to become more deterministic (i.e., the probability distribution of the mapping matrix becomes sharper), which reduces the likelihood of points being incorrectly mapped.

Therefore, we are the first to introduce entropy loss into this field, using it to regularize the entropy of the point-wise mapping matrix based on feature similarity, thereby reducing the uncertainty in the mapping. The entropy for a given row is defined as:

$$H_i = - \sum_j \Pi_{MN}(i, j) \log(\Pi_{MN}(i, j) + \epsilon), \quad (13)$$

where ϵ is a numerical stability term, set to 10^{-8} . The total entropy loss is expressed as:

$$L_{\text{entropy}} = \sum_i H_i. \quad (14)$$

2.3 Overall Losses

In deep functional map methods, the functional map is typically regularized using additional loss functions [Ren et al. 2019]. We simplify the structural constraints of the functional map matrix C

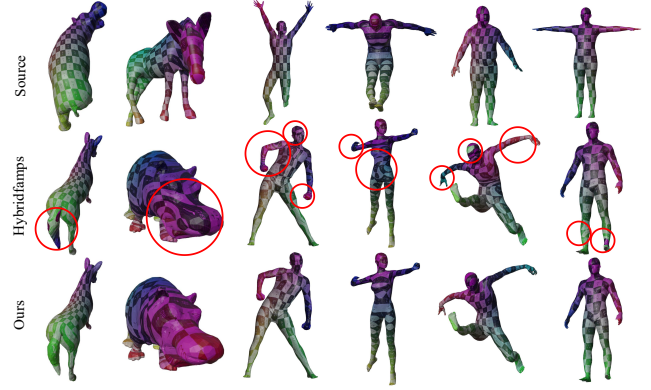


Figure 2: The first two columns show the qualitative results on the SMAL dataset, while the last four columns present the quantitative results on SHREC'19 for separate training on FAUST and SCAPE datasets, comparing with HybridFMap. Our method generates higher-quality correspondences, especially in cross-dataset generalization tests. Incorrect matches are marked by red circles.

in the LBO basis to define an orthogonality loss:

$$L_{\text{or}} = \|\Phi_M - \Pi_{MN} \Phi_N C_{NM}^\top\|_F^2. \quad (15)$$

We impose a bijectivity loss on the functional map:

$$L_{\text{bi}} = \|\Phi_M - \Pi_{MN} \Phi_N C_{MN}\|_F^2. \quad (16)$$

The L_{LB} loss is then defined as:

$$L_{\text{LB}} = \lambda_{\text{or}} L_{\text{or}} + \lambda_{\text{bi}} L_{\text{bi}}. \quad (17)$$

Since the LBO loss provides strong isometric supervision, we mainly rely on Eq. 17 in early training to prevent Eq. 11 and Eq. 14 from dominating. To balance them, we apply a linear annealing strategy (μ) defined as:

$$L_{\Pi} = \mu_1 \lambda_{\text{self-map}} L_{\text{self-map}} + \mu_2 \lambda_{\text{entropy}} L_{\text{entropy}}. \quad (18)$$

For the elastic basis loss and coupling loss [Cao et al. 2023], we follow the settings proposed by Bastian et al. [Bastian et al. 2024]. The total loss is as follows:

$$L_{\text{total}} = L_{\text{LB}} + L_{\Pi} + \mu L_{\text{Elas}} + \lambda_{\text{couple}} L_{\text{couple}}. \quad (19)$$

3 Experimental Results

We evaluate our method on near-isometric shape matching using the remeshed FAUST [Bogo et al. 2014], SCAPE [Anguelov et al. 2005], and SHREC'19 [Melzi et al. 2019] datasets, with remeshing providing a greater challenge. For non-isometric matching, we use the SMAL [Zuffi et al. 2017] and DeformingThings4D [Li et al. 2021] datasets. The DT4D-H dataset [Magnet et al. 2022], introduced by Magnet et al., poses challenges due to non-isometric deformations across categories. We also test robustness to topological noise using the TOPKIDS dataset [Löhner et al. 2016], which contains 26 synthetic child shapes created by merging intersecting parts.

We use the mean geodesic error (MGE) [Kim et al. 2011] as the evaluation metric. The results, shown in Tab. 1, demonstrate that our method outperforms state-of-the-art approaches in most settings, exhibiting better cross-dataset generalization. Fig. 2 presents a quantitative visual comparison with HybridFmap on selected datasets.

4 Ablation Studies

This section tests each component of our method for effectiveness using challenging non-isometric datasets, including SMAL and SHREC'19.

Our findings, presented in Tab. 2, show that relying solely on hybrid basis reconstruction does not generalize well for cross-dataset matching. Comparing the first three rows, the proposed loss terms improve model generalization and slightly enhance matching performance. The last row demonstrates that combining hybrid basis reconstruction with regularization terms improves performance while reducing functional map structural constraints.

Table 2: Ablation study conducted on SMAL and SHREC'19.

The first row represents the use of hybrid basis reconstruction and the L_{LB} loss with reduced structural constraints. The second row indicates the absence of the $L_{\text{self-map}}$. The third row represents the absence of the L_{entropy} . The last row corresponds to our complete method.

Geo. error ($\times 100$)	SMAL	SHREC'19
w/o L_{entropy} and $L_{\text{self-map}}$	3.3	4.6
w/o $L_{\text{self-map}}$	2.8	3.4
w/o L_{entropy}	3.0	3.4
Ours	2.5	3.1

5 Runtime Comparison and Discussion

We provide detailed runtime comparisons with HybridFmap [Bastian et al. 2024], together with additional experiments and extended discussions, in the supplementary material.

6 Limitations and Conclusion

We analyze the limitations of current state-of-the-art methods [Bastian et al. 2024] and propose a novel approach to address the challenges of solving the $k^2 \times k^2$ system in hybrid basis frameworks and training issues with the elastic basis. Additionally, we introduce entropy loss and self-map loss based on feature similarity for point-wise maps. Our method achieves state-of-the-art performance across multiple benchmark datasets. However, our approach has some limitations, including a space-for-time trade-off that slightly increases memory overhead. Meanwhile, addressing partial shapes or noisy point clouds under non-orthogonal basis functions remains an important open problem. Overall, our method performs excellently in various scenarios, marking a significant step forward in applying hybrid basis frameworks to real-world settings.

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